Lecture 9

The canonical pair: Weyl's relation and Stone–von Neumann

Objectives. *We prove the Stone–von Neumann theorem: every irreducible Weyl pair of strongly continuous unitary groups is unitarily equivalent to the pair on the line. The proof builds a nonzero projection from Gaussian integrals. The box shows why the commutator on a dense domain is insufficient.*

- Derive Weyl's relation from translations and phase multiplication.
- Construct the unitary equivalence from the Gaussian projection.
- Explain the roles of strong continuity, Weyl's relation and irreducibility.

Lecture 9, first part

Weyl's relation and the Stone–von Neumann theorem

9.1 Translations and Weyl's relation

Lecture 4 postulated that a particle on the line has the state space $L^2(\mathbb{R})$, with Q and P (Postulate 4.5). In this lecture we ask whether any other realisation of the canonical pair exists. The relation $QP - PQ = \mathrm{i}\hbar$ on a dense domain does not decide this, as the box shows below. Weyl's relation between translations does, together with strong continuity and irreducibility, at a fixed $\hbar$ and up to a unitary map. The answer matters for every later model of a particle on the line. If the particle has no further degrees of freedom and its translations meet these conditions, then its states, its Q and its P are those of $L^2(\mathbb{R})$, up to a unitary map. So the potentials of Lecture 10 and Lecture 11 change only the Hamiltonian, and the pair stays.

On the line the momentum moves states. We write $U(a)$, with its argument in parentheses, for a translation by the distance a, and keep U_t, with a subscript, for the evolution.

Proposition 9.1 (Momentum generates translations)**.** *For real a and $\psi \in L^2(\mathbb{R})$, put $(U(a)\psi)(x) = \psi(x-a)$. Then $(U(a))_{a\in\mathbb{R}}$ is a strongly continuous unitary group with the generator $P/\hbar$. So $U(a) = e^{-\mathrm{i}aP/\hbar}$, and $a \mapsto U(a)\psi$ is differentiable exactly when $\psi \in D(P)$.*

Proof. The substitution $y = x - a$ shows that $U(a)$ keeps norms, and $U(-a)$ is its inverse, so $U(a)$ is unitary. By the self-check of § 4.1, the transform of $U(a)\psi$ is $e^{-\mathrm{i}pa/\hbar}\tilde{\psi}$ for $\psi \in \mathcal{S}(\mathbb{R})$. So $U(a)$ agrees with $F^{-1}M_{e^{-\mathrm{i}ap/\hbar}}F$ on $\mathcal{S}(\mathbb{R})$, which is dense by the standing fact S4. Both are bounded, so they agree on every vector, as at the end of the proof of Proposition 7.9. By the multiplication case of Theorem 7.4, with the function p and the unitary F, this is a strongly

continuous unitary group, and its generator is $F^{-1}M_pF/\hbar = P/\hbar$. The differentiability claim is Proposition 7.3. ∎

So $D(P)$ is the set of states whose translations are differentiable in a, and for $\psi \in \mathcal{S}(\mathbb{R})$ the generator acts as $\mathrm{i}\,\mathrm{d}\psi(x-a)/\mathrm{d}a$ at $a=0$, which is $-\mathrm{i}\psi' = P\psi/\hbar$.

The position generates a second group. For real b, let $V(b)$ multiply by the phase $e^{\mathrm{i}bx/\hbar}$. We keep its argument, since V alone will be a potential from Lecture 10 on. By the multiplication case of Theorem 7.4, with the function $-x$ and the unitary $\mathbb{1}$, the operators $V(b)$ form a strongly continuous unitary group with the generator $-Q/\hbar$, so $V(b) = e^{\mathrm{i}bQ/\hbar}$. The translation $U(a)$ moves the position by a, and $V(b)$ moves the momentum by b. Indeed, by the self-check of § 4.1, the transform of $V(b)\psi$ is $\tilde{\psi}(p-b)$ for $\psi \in \mathcal{S}(\mathbb{R})$.

Proposition 9.2 (Weyl's relation)**.** *For all real a and b,*

$$U(a)V(b) = e^{-\mathrm{i}ab/\hbar}\,V(b)U(a).$$

Proof. For $\psi \in L^2(\mathbb{R})$, $(U(a)V(b)\psi)(x) = e^{\mathrm{i}b(x-a)/\hbar}\psi(x-a)$, and $(V(b)U(a)\psi)(x) = e^{\mathrm{i}bx/\hbar}\psi(x-a)$. So the first is $e^{-\mathrm{i}ab/\hbar}$ times the second. ∎

In *Quantum Mechanics I*, Thm 9.2, the clock and the shift on $\mathbb{C}^N$ obey $DT = \zeta TD$. That lecture promised the form of Weyl's relation that this course proves, and Proposition 9.2 is that form. Both translations together move a state through phase space. Interchanging their order multiplies the vector by $e^{-\mathrm{i}ab/\hbar}$.

Definition 9.3 (Weyl pair)**.** A *Weyl pair* on a Hilbert space $\mathcal{H} \neq \{0\}$ is two strongly continuous unitary groups $(U'(a))_{a\in\mathbb{R}}$ and $(V'(b))_{b\in\mathbb{R}}$ on $\mathcal{H}$ with

$$U'(a)V'(b) = e^{-\mathrm{i}ab/\hbar}\,V'(b)U'(a) \quad \text{for all real } a \text{ and } b.$$

Its *Weyl operators* are $W'(q,p) = e^{-\mathrm{i}qp/2\hbar}V'(p)U'(q)$. The pair is *irreducible* if the only closed subspaces of $\mathcal{H}$ that every $U'(a)$ and every $V'(b)$ map into themselves are $\{0\}$ and $\mathcal{H}$.

By Propositions 9.1 and 9.2 and the paragraph between them, the groups $U(a)$ and $V(b)$ of the line form a Weyl pair, the *Schrödinger pair*, with the Weyl operators $W(q,p) = e^{-\mathrm{i}qp/2\hbar}V(p)U(q)$, that is,

$$(W(q,p)\psi)(x) = e^{-\mathrm{i}qp/2\hbar}\,e^{\mathrm{i}px/\hbar}\,\psi(x-q).$$

A prime marks the operators of a general pair.

Two copies of the Schrödinger pair give a Weyl pair that is not irreducible. Let $\mathcal{H} = L^2(\mathbb{R}) \oplus L^2(\mathbb{R})$, the pairs (ψ_0, ψ_1) of functions, and let $U'(a)$ and $V'(b)$ act on both functions as $U(a)$ and $V(b)$ do. These are strongly continuous unitary groups. They satisfy Weyl's relation with the same $\hbar$, since it holds for each of the two functions. The pairs $(\psi_0, 0)$ form a closed subspace that is neither $\{0\}$ nor $\mathcal{H}$, and every $U'(a)$ and every $V'(b)$ maps it into itself. So this Weyl pair is not irreducible. A particle with spin $\frac{1}{2}$ on the line has a pair of this kind. Its states are pairs (ψ_0, ψ_1) of functions, and its translations act on both.

We write $z = (q,p)$ for a point of phase space, and

$$\sigma(z,w) = q'p - qp' \qquad \text{for } z = (q,p) \text{ and } w = (q',p').$$

The phase $\sigma(z,w)$ is linear in z and in w, and $\sigma(w,z) = -\sigma(z,w)$. Its absolute value is the area of the parallelogram with the sides z and w in the plane of (q,p), and its sign depends

on the order of the two points. So $\sigma(z,w)$ is an oriented area. It has two arguments, unlike the spectrum $\sigma(A)$ of an operator.

Proposition 9.4 (The product law). *Let $(U'(a))$ and $(V'(b))$ be a Weyl pair. Each $W'(z)$ is unitary, $z \mapsto W'(z)\psi$ is continuous for every $\psi \in \mathcal{H}$, and for all points z and w of phase space*

$$W'(z)W'(w) = e^{\mathrm{i}\sigma(z,w)/2\hbar}\, W'(z+w).$$

In particular, $W'(z)^\dagger = W'(-z)$, $W'(q,0) = U'(q)$ and $W'(0,p) = V'(p)$.

Proof. A product of unitaries is unitary. Let $z_n \to z$, with $z_n = (q_n, p_n)$. Then

$$\|V'(p_n)U'(q_n)\psi - V'(p)U'(q)\psi\| \le \|U'(q_n)\psi - U'(q)\psi\| + \|V'(p_n)U'(q)\psi - V'(p)U'(q)\psi\|\,,$$

since $V'(p_n)$ keeps norms, and both terms tend to 0. For the product, Weyl's relation with $a = q$ and $b = p'$ moves $U'(q)$ past $V'(p')$:

$$V'(p)U'(q)V'(p')U'(q') = e^{-\mathrm{i}qp'/\hbar}\, V'(p+p')U'(q+q').$$

With the phases of the three Weyl operators, the exponent is $\mathrm{i}/\hbar$ times

$$-\frac{qp+q'p'}{2} - qp' + \frac{(q+q')(p+p')}{2} = \frac{q'p-qp'}{2} = \frac{\sigma(z,w)}{2}.$$

Finally, $\sigma(z,-z) = 0$ gives $W'(z)W'(-z) = W'(0) = \mathbb{1}$, so $W'(z)^\dagger = W'(z)^{-1} = W'(-z)$. ∎

Figure 9.1 draws the product law on the line for $z = (s,0)$ and $w = (0,\hbar/s)$, with a length $s > 0$. Then $W(z) = U(s)$, $W(w) = V(\hbar/s)$ and $\sigma(z,w) = -\hbar$. So $W(z)W(w) = e^{-\mathrm{i}/2}W(z+w)$ and $W(w)W(z) = e^{\mathrm{i}/2}W(z+w)$. In each product, the exponent of the phase is $\mathrm{i}/2\hbar$ times the oriented area $\sigma(z,w)$ or $\sigma(w,z)$. The two products differ by $e^{-\mathrm{i}}$, whose exponent is $\mathrm{i}/\hbar$ times the whole oriented area $\sigma(z,w)$. This is the phase $e^{-\mathrm{i}ab/\hbar}$ of Weyl's relation with $a = s$ and $b = \hbar/s$.

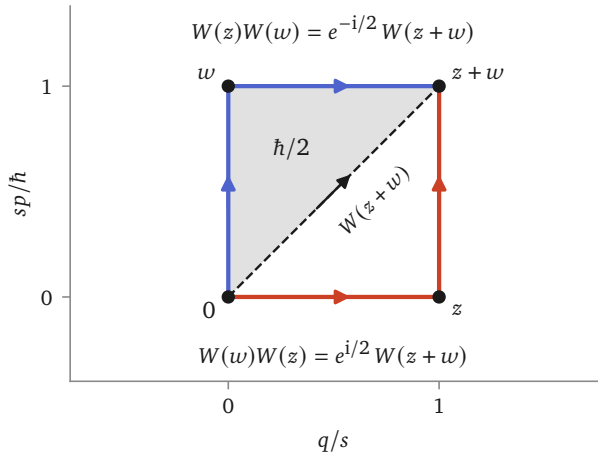


Figure 9.1: The plane of (q, p) in the units q/s and $sp/\hbar$, with the points $z = (s, 0)$ and $w = (0, \hbar/s)$. The Weyl operator $W(q', p')$ adds q' to the mean position and p' to the mean momentum. Each path follows the means of a state whose means are 0 as the two operators of one product act on it, the rightmost first. The upper path, through w, belongs to $W(z)W(w)$, and the lower path, through z, to $W(w)W(z)$. Both end at $z + w$, and the formula beside each path compares its product with $W(z + w)$, whose path is the dashed diagonal. The shaded triangle between the upper path and the diagonal has the area $\hbar/2$, and so has the triangle between the diagonal and the lower path. So each product carries half the area of the square: the exponent of its phase is $i/\hbar$ times $\sigma(z, w)/2 = -\hbar/2$ or $\sigma(w, z)/2 = \hbar/2$. The two products differ by the whole square, of area $\hbar$, through the phase e^{-i}. A phase that multiplies a whole vector changes no reading of that vector. The phases matter when such vectors are added.

For each z, the operators $W'(tz)$ form a unitary group in t, since $\sigma(tz, t'z) = 0$. For $\psi \in \mathcal{S}(\mathbb{R})$, the derivative in t of $(W(tq, tp)\psi)(x)$ at $t = 0$ is $(i/\hbar)(px\psi(x) - q(P\psi)(x))$. This is why $W(q, p)$ is often written $e^{i(pQ-qP)/\hbar}$. We use only the definition.

One matrix element will matter. Let $s > 0$ be a length. Every s is the oscillator length of Lecture 8 for the angular frequency $\nu = \hbar/ms^2$, and it is not the width s of Lecture 4. With $p_0 = p$, Theorem 8.10 says that $W(q, p)h_0$ is the coherent state ψ_α with $\alpha = (q/s + isp/\hbar)/\sqrt{2}$, where h_0 is the ground state $h_0(x) = (\pi s^2)^{-1/4}e^{-x^2/2s^2}$ of an oscillator with the length s (Lecture 8). By its series and the orthonormality of the h_n, $\langle h_0, \psi_\alpha\rangle = e^{-|\alpha|^2/2}$. So, with $g(q, p) = q^2/s^2 + s^2p^2/\hbar^2$,

$$\langle h_0, W(z)h_0\rangle = e^{-g(z)/4} \quad \text{for every } z. \tag{9.1}$$

Self-check 9.1

1. Let $\psi \in \mathcal{S}(\mathbb{R})$ be a unit vector. Show that the mean position in $U(a)\psi$ is $\langle Q\rangle + a$.

9.2 The theorem, and why $[Q, P] = i\hbar$ is not enough

Theorem 9.5 (Stone–von Neumann). *Let $(U'(a))$ and $(V'(b))$ be an irreducible Weyl pair on $\mathcal{H}$. Then there is a unitary T from $\mathcal{H}$ onto $L^2(\mathbb{R})$ with*

$$TU'(a) = U(a)T \quad \text{and} \quad TV'(b) = V(b)T \quad \text{for all real } a \text{ and } b.$$

§ 9.5 proves it. The unitary T carries the generators along. If $(U'(h)\psi - \psi)/h$ converges, then so does $(U(h)T\psi - T\psi)/h = T(U'(h)\psi - \psi)/h$, and conversely, by T^{-1}. So T maps the domain of the generator of $(U'(a))$ onto $D(P)$, and that generator to $P/\hbar$ (Definition 7.2). In the same way it carries the generator of $(V'(b))$ to $-Q/\hbar$. In words, an irreducible Weyl pair is the position and the momentum of a particle on the line, with their domains, up to a unitary change of the state space. *Quantum Mechanics I*, Exercise Sheet 9, proves the finite form: two unitaries on $\mathbb{C}^N$ that obey the relations of the clock and the shift, Weyl's relation and N-th powers equal to $\mathbb{1}$, are the clock and the shift in another basis.

The groups contain information that the commutator does not. The relation $QP - PQ = \mathrm{i}\hbar$ also holds in a box, and there it decides nothing. In a box, a translation must say what happens at the walls. For $\theta \in [0, 2\pi)$, extend $\psi \in L^2(0, L)$ to the line by $\psi(x + L) = e^{\mathrm{i}\theta}\psi(x)$, and let $(U_\theta(a)\psi)(x) = \psi(x - a)$ for $0 < x < L$. What leaves through the right wall enters at the left, multiplied by $e^{-\mathrm{i}\theta}$. So θ is the rule by which a translation crosses the walls, as Lecture 3 announced. Figure 9.2 shows these translations for one bump, at $\theta = 0$ and at $\theta = \pi$.

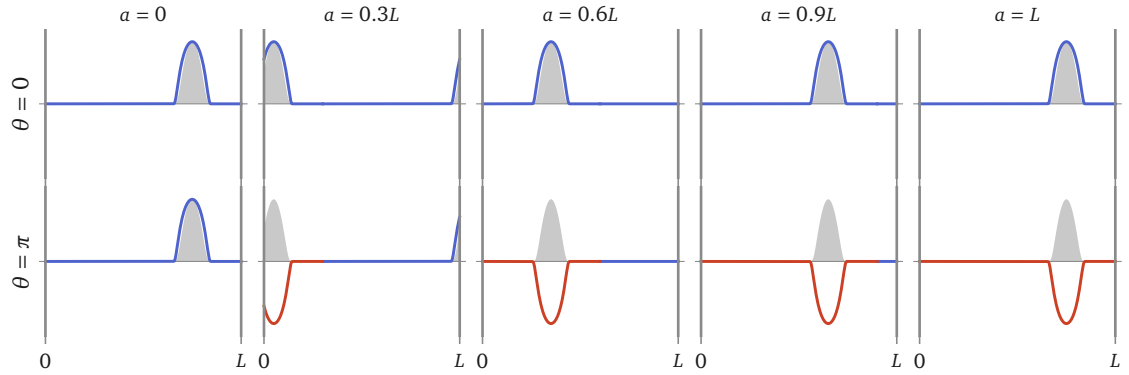


Figure 9.2: The translations $U_\theta(a)$ of one normalised smooth bump ψ in the box, which vanishes outside $[0.65L, 0.85L]$, for $\theta = 0$ (top row) and $\theta = \pi$ (bottom row). At these two values of θ the function $U_\theta(a)\psi$ is real. The curve shows it, scaled to the height 1, in blue where the phase factor of the extension is 1 and in red where it is -1. The grey area shows the density $|U_\theta(a)\psi|^2$, scaled to the same height, and the two rows have the same density. The part that crosses the right wall enters at the left, multiplied by $e^{-\mathrm{i}\theta}$. At $a = L$ the bump is back in its place, multiplied by $e^{-\mathrm{i}\theta}$, as $U_\theta(L) = e^{-\mathrm{i}\theta}\mathbb{1}$ says (Example 9.6). This is the group generated by $P_\theta/\hbar$. It is not the motion of a particle under the box Hamiltonian.

Example 9.6 (The commutator holds in the box, and Weyl's relation fails). Take the translations $U_\theta(a)$ on $L^2(0, L)$. We test them against the position in the box.

1. *The translations form a group.* Since $|\psi|$ extends with the period L, $U_\theta(a)$ keeps norms. The eigenvector $e_{\theta,k}$ of Theorem 3.10 is its own extension, and $e_{\theta,k}(x - a) = e^{-\mathrm{i}a(\theta+2\pi k)/L}e_{\theta,k}(x)$. So $U_\theta(a)$ agrees on the basis $(e_{\theta,k})$, complete by the standing fact S2, with the group of the diagonal operator with these eigenvectors and the eigenvalues $\hbar(\theta + 2\pi k)/L$, which is unitary by Theorem 7.4. Both are linear, so they agree on the finite combinations of the basis, which are dense. Both keep norms, so they agree on every vector, as at the end of the proof of Proposition 7.9. Hence $(U_\theta(a))$ is a strongly continuous unitary group, and by Theorem 7.4 its generator is

that diagonal operator divided by $\hbar$. By Theorem 3.10, the diagonal operator is P_θ.

2. *The position group.* In the box, Q multiplies by x, and it is bounded, with $\|Q\psi\| \le L\,\|\psi\|$. Let $V(b)$ multiply by $e^{\mathrm{i}bx/\hbar}$ on $L^2(0,L)$. Since $|x| \le L$ there, the bounds (7.2) give $(V(b)\psi - \psi)/b \to \mathrm{i}Q\psi/\hbar$ in norm for every ψ, so $(V(b))$ is a strongly continuous unitary group with the generator $-Q/\hbar$.
3. *The commutator holds.* The smooth functions that vanish near both walls form a dense subspace $C_c^\infty(0,L)$ (S4). The operators Q and P_θ map it into itself, and on it $QP_\theta\psi - P_\theta Q\psi = \mathrm{i}\hbar\psi$, by the product rule.
4. *Weyl's relation fails.* By the definition of the extension,
$$(U_\theta(L)\psi)(x) = \psi(x-L) = e^{-\mathrm{i}\theta}\psi(x),$$
so $U_\theta(L) = e^{-\mathrm{i}\theta}\mathbb{1}$. Weyl's relation at $a = L$ would then read $e^{-\mathrm{i}\theta}V(b) = e^{-\mathrm{i}Lb/\hbar}e^{-\mathrm{i}\theta}V(b)$, that is, $e^{-\mathrm{i}Lb/\hbar} = 1$ for every b, which is false for $b = \pi\hbar/L$.
5. *The pair is not the line's.* Nor is the box's pair unitarily equivalent to the line's. A unitary from $L^2(0,L)$ onto $L^2(\mathbb{R})$ that carried the box's Q to the line's would make the line's Q bounded and defined on every vector, and it is not. The function $(1+|x|)^{-1}$ is square-integrable, but it lies outside its domain.

So the commutator on a dense domain does not fix the pair. The theorem needs the groups.

Self-check 9.2

1. Show that $U_\theta(L) = e^{-\mathrm{i}\theta}\mathbb{1}$ from the eigenvalues of P_θ.

9.3 What the theorem decides

The theorem answers the question of this lecture. A particle on the line without further degrees of freedom has the state space $L^2(\mathbb{R})$, with Q and P on their domains of Theorem 4.6. This holds as soon as its translations in position and in momentum are strongly continuous, obey Weyl's relation and act irreducibly. No other realisation exists, up to a unitary map. The second part of this lecture proves the theorem.

Besides the continuity of the two groups, it assumes three things. Weyl's relation, not the commutator on a dense domain, as the box shows. Irreducibility, which excludes a spin. One value of $\hbar$. It decides nothing about the Hamiltonian. The free particle and the oscillator share the pair and differ in their energies.

The physics

A force on an oscillator acts through a Weyl operator. Take the oscillator of Lecture 8 with the Hamiltonian H and the evolution U_t, and add a force $F(t)$ that does not depend on the position. In this model the Hamiltonian is $H - F(t)Q$. The evolution from time 0 to time t is then, up to a phase, $U_t W(z(t))$, with
$$z(t) = \Big(-\frac{1}{m\nu}\int_0^t F(\tau)\sin\nu\tau\,\mathrm{d}\tau,\ \int_0^t F(\tau)\cos\nu\tau\,\mathrm{d}\tau\Big).$$
We state this without proof, since the course does not treat a Hamiltonian that depends on time. Take the force $F_0\cos\nu t$ at the frequency of the trap, and stop after a whole number of periods. Then $z(t) = (0, F_0 t/2)$, and U_t is a multiple of $\mathbb{1}$. So in this model the force takes h_0 to a multiple of the coherent state $W(0,F_0t/2)h_0$, as in the experiment of

> Lecture 8. For a general force and time, the evolution is the whole product $U_t W(z(t))$, not $W(z)$ alone.

Lecture 9, second part

The proof by von Neumann's projection

9.4 Von Neumann's projection

The first part stated the Stone–von Neumann theorem and showed what it decides for a particle on the line. In this part we prove the theorem. From the Weyl operators alone we build a nonzero projection. For an irreducible pair, a unit vector in its range then gives the unitary map onto $L^2(\mathbb{R})$. At the end we prove that the Schrödinger pair is itself irreducible.

On the line, the projection $|h_0\rangle\langle h_0|$ onto the oscillator's ground state picks one direction out of $L^2(\mathbb{R})$, and the Weyl operators carry h_0 to vectors whose closed span is $L^2(\mathbb{R})$ (Proposition 8.11). Von Neumann's idea is to build this projection from the Weyl operators alone, so that it exists for every Weyl pair.

The proof of Theorem 9.5 carries out this idea. First, we average the Weyl operators over phase space with a Gaussian weight. This gives an operator E_G, and we prove that it is a nonzero orthogonal projection for every Weyl pair. For a pair that is not irreducible, its range can have more than one dimension.

Next, we take a unit vector χ in the range of E_G. We call the set of all vectors $W'(z)\chi$ the *orbit* of χ, and the set of all vectors $W(z)h_0$ the orbit of h_0. This is not the classical orbit of Lecture 8, which is a path of means. Two vectors of the first orbit have the same inner product as the two vectors of the second orbit at the same points. So a finite combination of the first orbit and the same combination of the second have equal norms. Hence the rule $W'(z)\chi \mapsto W(z)h_0$ gives a linear map T on the finite combinations, and T keeps norms.

Finally, two density facts make T a unitary from all of $\mathcal{H}$ onto all of $L^2(\mathbb{R})$. The finite combinations of the first orbit are dense in $\mathcal{H}$ because the pair is irreducible. Those of the second are dense in $L^2(\mathbb{R})$ by Proposition 8.11.

Fix the length $s > 0$ and the function g of § 9.1. We use integrals over phase space $\mathbb{R}^2$ of continuous functions bounded by $Ce^{-cg(z)}$ for constants C and $c > 0$, and over $\mathbb{R}^4$ of continuous functions bounded by $Ce^{-c(g(v)+g(z))}$. These are limits of integrals over squares, as in calculus. On a square, a continuous function can be integrated in either order, and the bound makes the part outside a large square small. So the integrals over $\mathbb{R}^4$ can be taken in either order, and a shift of the variables or the reflection $z \mapsto -z$ does not change any of them. Example 9.9 also has one over $\mathbb{R}^3$, whose integrand, for $\varphi, \psi \in \mathcal{S}(\mathbb{R})$, has the modulus $e^{-g(z)/4}\,|\varphi(x)\psi(x-q)|$, and it can be integrated in any order for the same reason. For a Weyl pair, and vectors φ and ψ, the function $z \mapsto \langle \varphi, W'(z)\psi\rangle$ is continuous by Proposition 9.4, and bounded by $\|\varphi\|\,\|\psi\|$. Since $\int e^{-q^2/4s^2}\,\mathrm{d}q = 2s\sqrt{\pi}$ and $\int e^{-s^2p^2/4\hbar^2}\,\mathrm{d}p = 2\hbar\sqrt{\pi}/s$, by Lemma 8.8 at $k = 0$,

$$\int e^{-g(z)/4}\,\mathrm{d}z = 4\pi\hbar. \tag{9.2}$$

We return to the two orbits of the plan above. Figure 9.3 shows three vectors of each orbit.

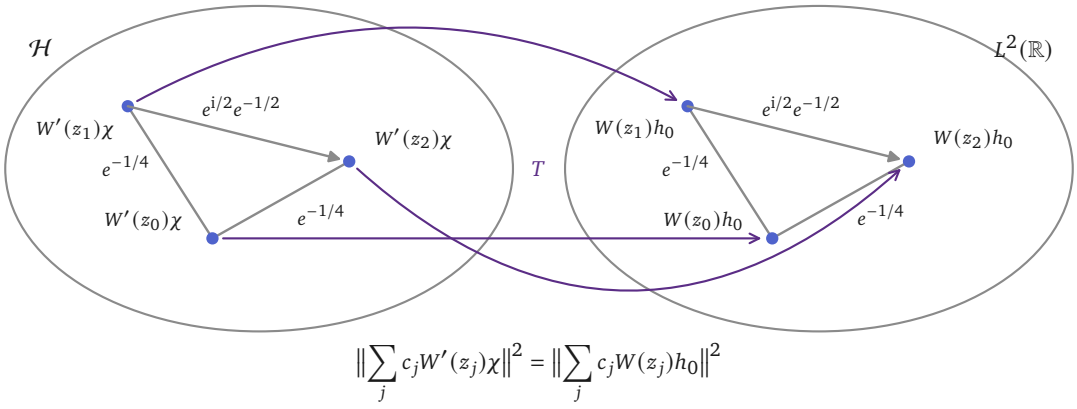


Figure 9.3: Three vectors of each orbit, at the points $z_0 = (0,0)$, $z_1 = (s,0)$ and $z_2 = (0,\hbar/s)$ of phase space. Left: the vectors $W'(z_j)\chi$ in $\mathcal{H}$. Right: the vectors $W(z_j)h_0$ in $L^2(\mathbb{R})$. The number on a side is the inner product of the two vectors at its ends. For the vectors at the points z and w, with the one at z in the first (left) slot, it is $e^{i\sigma(w,z)/2\hbar}e^{-g(w-z)/4}$ in both spaces. So these vectors are not orthogonal, and they are not a basis. On the side with the arrow head, the number is the inner product of the vector at the tail with the vector at the head. It is complex because the area is oriented: $\sigma(z_2,z_1) = \hbar$ and $\sigma(z_1,z_2) = -\hbar$. The drawn lengths are the distances between the vectors, at one scale, so the two triangles agree. The arrows between them are the map T. The formula at the foot of the figure holds for all finite combinations, and it makes T well defined. The ovals stand for the whole spaces, which are the closed spans of the two orbits when the pair is irreducible.

An operator can be given by its matrix elements. The key fact is the Riesz–Fréchet theorem, which *Linear Algebra*, Thm 13.40, cites. We prove what we need from Lemma 6.5.

Lemma 9.7 (An operator from its matrix elements)**.** *Let $B(\varphi,\psi)$ be a number for all $\varphi,\psi \in \mathcal{H}$, linear in ψ and antilinear in φ, with $|B(\varphi,\psi)| \le C\,\|\varphi\|\,\|\psi\|$. Then there is exactly one bounded operator A with $\langle\varphi, A\psi\rangle = B(\varphi,\psi)$ for all φ and ψ.*

Proof. Fix ψ, and let $\ell(\varphi) = \overline{B(\varphi,\psi)}$, which is linear in φ. Its kernel N is a subspace, and closed, since $|\ell(\varphi) - \ell(\varphi_n)| \le C\,\|\varphi - \varphi_n\|\,\|\psi\|$. If $\ell = 0$, put $A\psi = 0$. Otherwise $N \neq \mathcal{H}$, and by Lemma 6.5 there is a unit vector e orthogonal to N. For every φ, the vector $\ell(\varphi)e - \ell(e)\varphi$ lies in N, so it is orthogonal to e, and $\ell(\varphi) = \ell(e)\,\langle e,\varphi\rangle$. Put $A\psi = \overline{\ell(e)}\,e$. Then $\langle\varphi, A\psi\rangle = \overline{\langle A\psi,\varphi\rangle} = \overline{\ell(\varphi)} = B(\varphi,\psi)$. A vector orthogonal to every φ is 0, so $A\psi$ is the only such vector, that is, the only vector ξ with $\langle\varphi,\xi\rangle = B(\varphi,\psi)$ for every φ. Hence A is linear, since B is, and $\|A\psi\|^2 = B(A\psi,\psi) \le C\,\|A\psi\|\,\|\psi\|$ gives $\|A\psi\| \le C\,\|\psi\|$. ∎

Definition 9.8 (Von Neumann's operator)**.** For a Weyl pair on $\mathcal{H}$, the operator E_G is the bounded operator with

$$\langle\varphi, E_G\psi\rangle = \frac{1}{2\pi\hbar}\int e^{-g(z)/4}\,\langle\varphi, W'(z)\psi\rangle\,\mathrm{d}z \quad \text{for all } \varphi,\psi \in \mathcal{H}.$$

The right side is linear in ψ and antilinear in φ, and by (9.2) its modulus is at most $2\,\|\varphi\|\,\|\psi\|$. So Lemma 9.7 gives E_G. The letter G is for the Gaussian. On the line, by (9.1), the weight is $\langle h_0, W(z)h_0\rangle$, and we now compute E_G there directly.

Example 9.9 (The operator E_G on the line)**.** For the Schrödinger pair, $E_G = |h_0\rangle\langle h_0|$. To see this, let $\varphi, \psi \in \mathcal{S}(\mathbb{R})$. By the formula for $W(q,p)$,

$$\langle \varphi, E_G\psi\rangle = \frac{1}{2\pi\hbar}\iiint e^{-q^2/4s^2}e^{-s^2p^2/4\hbar^2}e^{\mathrm{i}p(x-q/2)/\hbar}\,\overline{\varphi(x)}\,\psi(x-q)\,\mathrm{d}x\,\mathrm{d}q\,\mathrm{d}p.$$

The integrand is continuous and decays fast in all three variables, so we may integrate over p first. Lemma 8.8, with $b = 2\hbar/s$, gives

$$\int e^{-s^2p^2/4\hbar^2}e^{\mathrm{i}p(x-q/2)/\hbar}\,\mathrm{d}p = \frac{2\hbar\sqrt{\pi}}{s}\,e^{-(x-q/2)^2/s^2}.$$

Now put $y = x - q$, so that $q = x - y$ and $x - q/2 = (x+y)/2$. The two remaining Gaussians combine to

$$e^{-(x-y)^2/4s^2}e^{-(x+y)^2/4s^2} = e^{-(x^2+y^2)/2s^2} = s\sqrt{\pi}\,h_0(x)h_0(y).$$

The constants give $(2\pi\hbar)^{-1}(2\hbar\sqrt{\pi}/s)(s\sqrt{\pi}) = 1$. Since h_0 is real,

$$\langle \varphi, E_G\psi\rangle = \iint \overline{\varphi(x)}\,h_0(x)h_0(y)\psi(y)\,\mathrm{d}x\,\mathrm{d}y = \langle \varphi, h_0\rangle\,\langle h_0, \psi\rangle\,.$$

Both sides are bounded in φ and in ψ, and $\mathcal{S}(\mathbb{R})$ is dense by the standing fact S4. So this holds for all φ and ψ in $L^2(\mathbb{R})$. The weight and the factor $1/2\pi\hbar$ give exactly the projection onto h_0.

Lemma 9.10 (Von Neumann's projection)**.** *The operator E_G is self-adjoint, and for every point w of phase space*

$$E_G W'(w) E_G = e^{-g(w)/4}\,E_G. \tag{9.3}$$

In particular $E_G^2 = E_G$, so E_G is an orthogonal projection.

Proof. First, by the definition, and since the weight is real,

$$\langle E_G\varphi, \psi\rangle = \overline{\langle \psi, E_G\varphi\rangle} = \frac{1}{2\pi\hbar}\int e^{-g(z)/4}\,\langle W'(z)\varphi, \psi\rangle\,\mathrm{d}z.$$

Since $\langle W'(z)\varphi, \psi\rangle = \langle \varphi, W'(-z)\psi\rangle$ and $g(-z) = g(z)$, the reflection $z \mapsto -z$ turns this into $\langle \varphi, E_G\psi\rangle$. So E_G is self-adjoint.

Next, we compute $E_G W'(w) E_G$. The product law turns it into one more average of the operators $W'(u)$ over phase space, with a new weight. We compute this weight and find that it is $e^{-g(w)/4}$ times the weight $e^{-g(u)/4}$ of E_G. Let $\varphi, \psi \in \mathcal{H}$. By the definition applied twice, and $W'(v)^\dagger = W'(-v)$ and $W'(w)^\dagger = W'(-w)$,

$$\langle \varphi, E_G W'(w) E_G\psi\rangle = \frac{1}{(2\pi\hbar)^2}\iint e^{-(g(v)+g(z))/4}\,\langle \varphi, W'(v)W'(w)W'(z)\psi\rangle\,\mathrm{d}v\,\mathrm{d}z.$$

By the product law, used twice, $W'(v)W'(w)W'(z)$ is $e^{\mathrm{i}[\sigma(w,z)+\sigma(v,w+z)]/2\hbar}W'(v+w+z)$. Substitute $z = u - w - v$ for fixed v. Since $\sigma(z,w)$ is linear in z and in w, and $\sigma(w,w) = 0$, the phase becomes $\sigma(w,u) + \sigma(v,u+w)$. The new integrand is still of our kind in (v,u). Indeed, $u = v + w + z$ gives $g(u) \le 3(g(v) + g(z) + g(w))$, so $g(v) + g(z) \ge (g(v) + g(u))/6 - g(w)/2$. Integrating over v first,

$$\langle \varphi, E_G W'(w) E_G\psi\rangle = \frac{1}{2\pi\hbar}\int K(u)\,\langle \varphi, W'(u)\psi\rangle\,\mathrm{d}u,$$

where, with $c = u - w$,

$$K(u) = \frac{e^{i\sigma(w,u)/2\hbar}}{2\pi\hbar} \int e^{-(g(v)+g(c-v))/4}\, e^{i\sigma(v,u+w)/2\hbar}\, dv.$$

Expanding the squares gives $g(v) + g(c - v) = 2g(v - c/2) + g(c)/2$. Put $v = c/2 + y$. Then $\sigma(c/2, u + w) = \frac{1}{2}\sigma(u - w, u + w) = \sigma(u, w)$, whose phase cancels $e^{i\sigma(w,u)/2\hbar}$. With $d = u + w = (d_q, d_p)$ and $y = (y_q, y_p)$, the phase $\sigma(y, d) = d_q y_p - y_q d_p$ splits, and Lemma 8.8, applied to each variable with $b = \sqrt{2}\, s$ and with $b = \sqrt{2}\,\hbar/s$, gives

$$\int e^{-g(y)/2} e^{i\sigma(y,d)/2\hbar}\, dy = 2\pi\hbar\, e^{-g(d)/8}.$$

So $K(u) = e^{-g(u-w)/8 - g(u+w)/8} = e^{-g(u)/4} e^{-g(w)/4}$, and the right side is $e^{-g(w)/4}\, \langle \varphi, E_G \psi \rangle$. This is (9.3). With $w = 0$, it reads $E_G^2 = E_G$. Then $E_G\psi$ is orthogonal to $\psi - E_G\psi$ for every ψ, since $\langle E_G\psi, \psi - E_G\psi \rangle = \langle \psi, E_G\psi - E_G^2\psi \rangle = 0$. ∎

On the line, Example 9.9 gives $E_G W(w) E_G = |h_0\rangle\langle h_0|\, W(w)\, |h_0\rangle\langle h_0| = \langle h_0, W(w) h_0 \rangle\, |h_0\rangle\langle h_0|$. So there (9.3) is (9.1), and the lemma says that every Weyl pair has the same identity.

A projection may be 0, and $E_G = 0$ would make the lemma empty. It is not.

Lemma 9.11 (The projection is not zero). *For every Weyl pair, $E_G \neq 0$.*

Proof. We test E_G on the vectors $W'(z)\psi$ and average over z with a wide Gaussian weight. Let ψ be a unit vector, and $f(u) = \langle \psi, W'(u)\psi \rangle$. Then f is continuous, $|f| \leq 1$ and $f(0) = 1$. For $\epsilon > 0$, consider

$$N_\epsilon = \frac{1}{2\pi\hbar} \int e^{-\epsilon g(z)}\, \langle W'(z)\psi, E_G W'(z)\psi \rangle\, dz.$$

If $E_G = 0$, then $N_\epsilon = 0$. By the product law, $W'(-z)W'(u)W'(z) = e^{i\sigma(u,z)/\hbar} W'(u)$. So by the definition of E_G,

$$N_\epsilon = \frac{1}{(2\pi\hbar)^2} \iint e^{-\epsilon g(z)} e^{-g(u)/4} e^{i\sigma(u,z)/\hbar} f(u)\, du\, dz.$$

For fixed u, Lemma 8.8 in each variable gives $\int e^{-\epsilon g(z)} e^{i\sigma(u,z)/\hbar} dz = (\pi\hbar/\epsilon) e^{-g(u)/4\epsilon}$. So $N_\epsilon = \int \gamma_\epsilon f\, du$, with the positive weight

$$\gamma_\epsilon(u) = \frac{1}{4\pi\hbar\epsilon}\, e^{-(1+\epsilon)g(u)/4\epsilon}, \qquad \int \gamma_\epsilon\, du = \frac{1}{1+\epsilon}, \qquad \int \gamma_\epsilon g\, du = \frac{4\epsilon}{(1+\epsilon)^2},$$

since the integrals behind (9.2) give $\int e^{-\lambda g(u)/4} du = 4\pi\hbar/\lambda$ for every $\lambda > 0$, and, by differentiating in λ, $\int g\, e^{-\lambda g(u)/4} du = 16\pi\hbar/\lambda^2$. As $\epsilon \to 0$, the weight concentrates at $u = 0$. Let $\eta > 0$, and choose $\delta > 0$ with $|f(u) - 1| < \eta$ when $g(u) \leq \delta$. Since $|f - 1| \leq 2$, this gives $|f(u) - 1| \leq \eta + 2g(u)/\delta$ for every u. So

$$|N_\epsilon - 1| \leq \int \gamma_\epsilon\, |f - 1|\, du + \frac{\epsilon}{1+\epsilon} \leq \eta + \frac{8\epsilon}{\delta} + \epsilon,$$

which is less than 2η for small ϵ. So $N_\epsilon \to 1$, and $E_G \neq 0$. ∎

The integral N_ϵ tends to 1 for every unit vector, whereas $E_G = 0$ would make every N_ϵ vanish.

Self-check 9.4

1. Use (9.3) to show that $\langle \chi, W'(z)\chi \rangle = e^{-g(z)/4}$ for every unit vector χ in the range of E_G.

9.5 Proof of the Stone–von Neumann theorem

Proof of Theorem 9.5.

1. *A vector.* By Lemma 9.11, the range of E_G contains a unit vector χ. By (9.3), $E_G\chi = \chi$ and the self-adjointness of E_G, $\langle \chi, W'(z)\chi\rangle = \langle \chi, E_G W'(z) E_G \chi\rangle = e^{-g(z)/4}$. By the product law, $W'(z)^\dagger W'(w) = W'(-z)W'(w) = e^{i\sigma(w,z)/2\hbar} W'(w-z)$, so

$$\langle W'(z)\chi, W'(w)\chi\rangle = e^{i\sigma(w,z)/2\hbar}\, e^{-g(w-z)/4} \quad \text{for all } z \text{ and } w. \tag{9.4}$$

2. *Its translates span $\mathcal{H}$.* Let $\mathcal{K}$ be the closure of the finite combinations of the vectors $W'(z)\chi$. By the product law, each $W'(w)$ maps $W'(z)\chi$ to a multiple of $W'(w+z)\chi$. So it maps the combinations into themselves, and, being bounded, it maps $\mathcal{K}$ into $\mathcal{K}$. In particular $U'(a) = W'(a,0)$ and $V'(b) = W'(0,b)$ do. Since $\chi \in \mathcal{K}$ and the pair is irreducible, $\mathcal{K} = \mathcal{H}$.
3. *The same on the line.* By (9.1) and the same product law, the vectors $W(z)h_0$ satisfy (9.4) with h_0 in place of χ. Their closed span contains every $W(q,0)h_0 = h_0(x-q)$. By Proposition 8.11, only 0 is orthogonal to all of these, so by Lemma 6.5 the closed span is $L^2(\mathbb{R})$.
4. *The unitary.* On a finite combination, put $T\big(\sum_j c_j W'(z_j)\chi\big) = \sum_j c_j W(z_j)h_0$. By the first and the third step, both sides have the same squared norm

$$\sum_{j,k} \overline{c_j} c_k\, e^{i\sigma(z_k,z_j)/2\hbar} e^{-g(z_k-z_j)/4}.$$

   Figure 9.3 shows this comparison on three vectors. So two combinations that are equal have equal images, since their difference has norm 0, and T is a well-defined linear map that keeps norms. If ξ_n are combinations with $\xi_n \to \xi$, the images form a Cauchy sequence, and $T\xi$ is their limit. The limit does not depend on the sequence, since $\|T\xi_n - T\xi_n'\| = \|\xi_n - \xi_n'\|$. This defines T on $\mathcal{H}$, by the second step, and T keeps norms. Its range is complete, so closed, and it contains every $W(z)h_0$, so by the third step it is $L^2(\mathbb{R})$. A linear map that keeps norms keeps inner products, by the polarisation identity. So $\langle T\xi, T\zeta\rangle = \langle \xi, \zeta\rangle$, and since T is onto, its inverse is its adjoint. So T is unitary.
5. *It carries one pair to the other.* By the product law,

$$TW'(w)W'(z)\chi = e^{i\sigma(w,z)/2\hbar}\, TW'(w+z)\chi = e^{i\sigma(w,z)/2\hbar}\, W(w+z)h_0,$$

   which is $W(w)W(z)h_0 = W(w)TW'(z)\chi$. So $TW'(w) = W(w)T$ on the combinations, and, since both sides are bounded, on $\mathcal{H}$. With $w = (a,0)$ and $w = (0,b)$, this is the claim. ∎

The range of E_G is the one-dimensional subspace spanned by χ. Indeed, let χ_1 be in the range and orthogonal to χ. Then, by (9.3),

$$\langle \chi_1, W'(z)\chi\rangle = \langle \chi_1, E_G W'(z) E_G \chi\rangle = e^{-g(z)/4}\,\langle \chi_1, \chi\rangle = 0 \quad \text{for every } z.$$

By the second step, χ_1 is orthogonal to all of $\mathcal{H}$, so $\chi_1 = 0$. So $E_G = |\chi\rangle\langle\chi|$, and T maps χ to h_0.

The Schrödinger pair is itself irreducible.

Proposition 9.12 (The Schrödinger pair is irreducible)**.** *The only closed subspaces of $L^2(\mathbb{R})$ that every $U(a)$ and every $V(b)$ map into themselves are $\{0\}$ and $L^2(\mathbb{R})$.*

Proof. Let $\mathcal{K} \neq \{0\}$ be such a subspace. Since $U(a)^\dagger = U(-a)$ and $V(b)^\dagger = V(-b)$, the restrictions to $\mathcal{K}$ are a Weyl pair on $\mathcal{K}$, with the restrictions of the $W(z)$ as its Weyl operators. By Lemmas 9.10 and 9.11 for that pair, there is a unit vector $\xi \in \mathcal{K}$ with $\langle \xi, W(z)\xi\rangle = e^{-g(z)/4}$ for every z, as in the first step of the proof above. Let E_G now be the operator of the whole pair on $L^2(\mathbb{R})$. Then $\langle \xi, E_G\xi\rangle = (2\pi\hbar)^{-1}\int e^{-g(z)/2}\mathrm{d}z = 1$, and since E_G is a projection, $\|\xi - E_G\xi\|^2 = 1 - \langle \xi, E_G\xi\rangle = 0$. So $E_G\xi = \xi$. By Example 9.9, $E_G = |h_0\rangle\langle h_0|$, so $\xi = \langle h_0, \xi\rangle\, h_0$, and $\langle h_0, \xi\rangle \neq 0$ since $\xi \neq 0$. So h_0 lies in $\mathcal{K}$. Then so does every $W(z)h_0$, and $\mathcal{K} = L^2(\mathbb{R})$ by the third step of the proof. ∎

Self-check 9.5

1. Let $\mathcal{H} = L^2(\mathbb{R}) \oplus L^2(\mathbb{R})$, the pairs (ψ_0, ψ_1), with $U'(a)$ and $V'(b)$ acting on both functions as $U(a)$ and $V(b)$ do. What is E_G? What is the dimension of its range?
2. Why does the proof need Lemma 9.11, when Lemma 9.10 already gives an orthogonal projection?
3. In the pair of the first check, take $\chi = (h_0, 0)$. What is the closed span of the orbit of χ? Which step of the proof of Theorem 9.5 fails?
4. Suppose that a finite combination $\sum_j c_j W'(z_j)\chi$ is 0. Why is $\sum_j c_j W(z_j)h_0$ also 0?
5. The map T keeps norms. Why is its range all of $L^2(\mathbb{R})$?

The canonical pair

States: for a particle on the line with no further degree of freedom, every irreducible Weyl pair is the line's, up to a unitary map (Theorem 9.5). So its state space is $L^2(\mathbb{R})$. *Observables:* the generators of the two groups are $P/\hbar$ and $-Q/\hbar$, with the domains of P and Q, and $U(a)V(b) = e^{-\mathrm{i}ab/\hbar}V(b)U(a)$ (Proposition 9.2). The relation $QP - PQ = \mathrm{i}\hbar$ on a dense domain does not fix them: in the box it holds and Weyl's relation fails. *Evolution:* the theorem says nothing about the Hamiltonian. *Measurement:* for an irreducible pair, von Neumann's projection E_G, built from the Weyl operators alone, is the projection onto one vector, which the unitary map carries to the oscillator's ground state h_0 (§ 9.5, after the proof of Theorem 9.5).

Summary.

- *Momentum generates translations*, $U(a) = e^{-\mathrm{i}aP/\hbar}$ (Proposition 9.1), and $V(b) = e^{\mathrm{i}bQ/\hbar}$; *Weyl's relation* $U(a)V(b) = e^{-\mathrm{i}ab/\hbar}V(b)U(a)$ (Proposition 9.2).
- *The product law* $W'(z)W'(w) = e^{\mathrm{i}\sigma(z,w)/2\hbar}W'(z+w)$ for every Weyl pair (Proposition 9.4).
- *Von Neumann's projection*: E_G is an orthogonal projection with $E_G W'(w) E_G = e^{-g(w)/4}E_G$, and $E_G \neq 0$ (Lemmas 9.10 and 9.11).
- *Stone–von Neumann*: every irreducible Weyl pair is unitarily equivalent to the Schrödinger pair (Theorem 9.5), which is irreducible (Proposition 9.12).

Lecture 10 adds a potential to the free energy: the pair stays, and the Hamiltonian changes.

Exercises. Exercise Sheet 9.

Self-check answers.

9.1 1. Substituting $y = x - a$, $\int x\,|\psi(x-a)|^2\,\mathrm{d}x = \int (y+a)\,|\psi(y)|^2\,\mathrm{d}y = \langle Q\rangle + a$, since $\|\psi\| = 1$.

9.2 1. On the eigenvector $e_{\theta,k}$, the group multiplies by $e^{-\mathrm{i}L(\theta+2\pi k)/L} = e^{-\mathrm{i}\theta}$, the same for every k. Two bounded operators that agree on a basis agree everywhere.

9.4 1. Since $E_G\chi = \chi$ and E_G is self-adjoint, $\langle \chi, W'(z)\chi\rangle = \langle E_G\chi, W'(z)E_G\chi\rangle = \langle \chi, E_G W'(z) E_G\chi\rangle = e^{-g(z)/4}\,\|\chi\|^2$.

9.5 1. The integrand acts on each function, so $E_G(\psi_0,\psi_1) = (\langle h_0,\psi_0\rangle\, h_0, \langle h_0,\psi_1\rangle\, h_0)$. Its range is spanned by $(h_0,0)$ and $(0,h_0)$: dimension 2. 2. The operator 0 is also an orthogonal projection, and it satisfies (9.3). The first step of the proof needs a unit vector in the range of E_G, and for $E_G = 0$ there is none. 3. Here $W'(z)\chi = (W(z)h_0, 0)$. So, by the third step of the proof, the closed span is the subspace of the pairs $(\psi_0, 0)$, which is not all of $\mathcal{H}$. The second step fails, since this pair is not irreducible. The map T is then defined only on this subspace. 4. By the first and the third step, the two combinations have the same squared norm, the double sum of the fourth step. So the second has norm 0. The argument uses no orthogonality and no linear independence of these vectors. 5. The range is complete, so closed, and it contains every $W(z)h_0$. By the third step, the closed span of these vectors is $L^2(\mathbb{R})$. Keeping norms alone gives only a closed subspace.

References

Other books prove some results of this lecture. Hall, Ex. 14.5, makes the point of Example 9.6 for $\theta = 0$ on the interval $[-1,1]$. Hall, Lemma 14.11, proves (9.3) by checking it on $L^2(\mathbb{R})$ first. Prugovečki, Lemma IV.6.3, proves $E_G \neq 0$ with Fourier series. Hall, Prop. 14.10, proves it with the Fourier transform of $\mathbb{R}^2$. Hall, Prop. 14.7, states Proposition 9.12.

Hall, *Quantum Theory for Mathematicians*, Ex. 14.5 (p. 285), Def. 14.6, Prop. 14.7 and Thm 14.8 (p. 286), Prop. 14.10 (pp. 288–289), Lemma 14.11 (pp. 289–290). Prugovečki, *Quantum Mechanics in Hilbert Space*, Thm IV.6.4 (p. 342), §IV.6.5 (pp. 342–347), Lemma IV.6.3 (p. 344). *Linear Algebra for Physics*, Thm 13.40. *Quantum Mechanics I*, Thm 9.2 and Exercise Sheet 9.